

Dokumentowe bazy danych – MongoDB

Ćwiczenie 2

Odtwórz z backupu bazę **north0**

```
mongorestore ...
```

```
use north0
```

Baza **north0** jest kopią relacyjnej bazy danych **Northwind**

- poszczególne kolekcje odpowiadają tabelom w oryginalnej bazie **Northwind**

zapoznaj się ze strukturą dokumentów w bazie **North0**

```
db.customers.find();
db.orders.find();
db.orderdetails.find();
```

produkty, kategorie, dostawcy

```
db.categories.find().limit(2);
```

- wynik

```
[
  {
    _id: { $oid: "67eb06a27b1ef941c950fb49" },
    CategoryID: 1,
    CategoryName: "Beverages",
    Description: "Soft drinks, coffees, teas, beers, and ales",
  },
  {
    _id: { $oid: "67eb06a27b1ef941c950fb4a" },
    CategoryID: 2,
    CategoryName: "Condiments",
    Description: "Sweet and savory sauces, relishes, spreads, and seasonings",
  },
];
```

```
db.products.find().limit(2);
```

- wynik

```
[
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f50" },
    CategoryID: 1,
```

```

    Discontinued: false,
    ProductID: 1,
    ProductName: "Chai",
    QuantityPerUnit: "10 boxes x 20 bags",
    ReorderLevel: 10,
    SupplierID: 1,
    UnitPrice: 18,
    UnitsInStock: 39,
    UnitsOnOrder: 0,
  },
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f51" },
    CategoryID: 1,
    Discontinued: false,
    ProductID: 2,
    ProductName: "Chang",
    QuantityPerUnit: "24 - 12 oz bottles",
    ReorderLevel: 25,
    SupplierID: 1,
    UnitPrice: 19,
    UnitsInStock: 17,
    UnitsOnOrder: 40,
  },
];

```

```
db.suppliers.find().limit(2);
```

- wynik

```

[
  {
    _id: { $oid: "63a05d3cbb3b972d6f4e09f7" },
    Address: "49 Gilbert St.",
    City: "London",
    CompanyName: "Exotic Liquids",
    ContactName: "Charlotte Cooper",
    ContactTitle: "Purchasing Manager",
    Country: "UK",
    Fax: null,
    HomePage: null,
    Phone: "(171) 555-2222",
    PostalCode: "EC1 4SD",
    Region: null,
    SupplierID: 1,
  },
  {
    _id: { $oid: "63a05d3cbb3b972d6f4e09f8" },
    Address: "P.O. Box 78934",
    City: "New Orleans",
    CompanyName: "New Orleans Cajun Delights",
    ContactName: "Shelley Burke",
    ContactTitle: "Order Administrator",
    Country: "USA",
    Fax: null,
    HomePage: "#CAJUN.HTM#",
    Phone: "(100) 555-4822",
    PostalCode: "70117",
    Region: "LA",
    SupplierID: 2,
  },
];

```

Przykład 1

```
db.products.aggregate([
  {
    $match: {},
  },

  //suppliers
  {
    $lookup: {
      from: "suppliers",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Supplier",
    },
  },
  {
    $limit: 2,
  },
]);
```

- pole **Supplier** jest tablicą
 - zawiera 1 obiekt
- wynik

```
[
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f50" },
    CategoryID: 1,
    Discontinued: false,
    ProductID: 1,
    ProductName: "Chai",
    QuantityPerUnit: "10 boxes x 20 bags",
    ReorderLevel: 10,
    Supplier: [
      {
        _id: { $oid: "63a05d3cbb3b972d6f4e09f7" },
        SupplierID: 1,
        CompanyName: "Exotic Liquids",
        ContactName: "Charlotte Cooper",
        ContactTitle: "Purchasing Manager",
        Address: "49 Gilbert St.",
        City: "London",
        Region: null,
        PostalCode: "EC1 4SD",
        Country: "UK",
        Phone: "(171) 555-2222",
        Fax: null,
        HomePage: null,
      },
    ],
    SupplierID: 1,
    UnitPrice: 18,
    UnitsInStock: 39,
    UnitsOnOrder: 0,
  },
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f51" },
    CategoryID: 1,
    Discontinued: false,
    ProductID: 2,
```

```

    ProductName: "Chang",
    QuantityPerUnit: "24 - 12 oz bottles",
    ReorderLevel: 25,
    Supplier: [
      {
        _id: { $oid: "63a05d3cbb3b972d6f4e09f7" },
        SupplierID: 1,
        CompanyName: "Exotic Liquids",
        ContactName: "Charlotte Cooper",
        ContactTitle: "Purchasing Manager",
        Address: "49 Gilbert St.",
        City: "London",
        Region: null,
        PostalCode: "EC1 4SD",
        Country: "UK",
        Phone: "(171) 555-2222",
        Fax: null,
        HomePage: null,
      },
    ],
    SupplierID: 1,
    UnitPrice: 19,
    UnitsInStock: 17,
    UnitsOnOrder: 40,
  },
];

```

- \$unwind
 - pole **Supplier** jest obiektem/dokumentem

```

db.products.aggregate([
  {
    $match: {},
  },

  //suppliers
  {
    $lookup: {
      from: "suppliers",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Supplier",
    },
  },
  {
    $unwind: "$Supplier",
  },
  {
    $limit: 2,
  },
]);

```

- wynik

```

[
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f50" },
    CategoryID: 1,
    Discontinued: false,
    ProductID: 1,
    ProductName: "Chai",
    QuantityPerUnit: "10 boxes x 20 bags",
    ReorderLevel: 10,
  }
]

```

```

Supplier: {
  _id: { $oid: "63a05d3cbb3b972d6f4e09f7" },
  SupplierID: 1,
  CompanyName: "Exotic Liquids",
  ContactName: "Charlotte Cooper",
  ContactTitle: "Purchasing Manager",
  Address: "49 Gilbert St.",
  City: "London",
  Region: null,
  PostalCode: "EC1 4SD",
  Country: "UK",
  Phone: "(171) 555-2222",
  Fax: null,
  HomePage: null,
},
SupplierID: 1,
UnitPrice: 18,
UnitsInStock: 39,
UnitsOnOrder: 0,
},
{
  _id: { $oid: "63a0607abb3b972d6f4e1f51" },
  CategoryID: 1,
  Discontinued: false,
  ProductID: 2,
  ProductName: "Chang",
  QuantityPerUnit: "24 - 12 oz bottles",
  ReorderLevel: 25,
  Supplier: {
    _id: { $oid: "63a05d3cbb3b972d6f4e09f7" },
    SupplierID: 1,
    CompanyName: "Exotic Liquids",
    ContactName: "Charlotte Cooper",
    ContactTitle: "Purchasing Manager",
    Address: "49 Gilbert St.",
    City: "London",
    Region: null,
    PostalCode: "EC1 4SD",
    Country: "UK",
    Phone: "(171) 555-2222",
    Fax: null,
    HomePage: null,
  },
  SupplierID: 1,
  UnitPrice: 19,
  UnitsInStock: 17,
  UnitsOnOrder: 40,
},
];

```

- projekcja
 - tylko wybrane pola z `Supplier`

```

db.products.aggregate([
  {
    $match: {},
  },

  //suppliers
  {
    $lookup: {
      from: "suppliers",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Supplier_tmp",
    },
  },
]);

```

```

    },
  },
  {
    $unwind: "$Supplier_tmp",
  },
  {
    $addFields: {
      Supplier: {
        SupplierID: "$SupplierID",
        CompanyName: "$Supplier_tmp.CompanyName",
        Country: "$Supplier_tmp.Country",
      },
    },
  },
  {
    $project: {
      SupplierID: 0,
      Supplier_tmp: 0,
    },
  },
  {
    $limit: 2,
  },
});

```

- wynik

```

[
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f50" },
    CategoryID: 1,
    Discontinued: false,
    ProductID: 1,
    ProductName: "Chai",
    QuantityPerUnit: "10 boxes x 20 bags",
    ReorderLevel: 10,
    Supplier: {
      SupplierID: 1,
      CompanyName: "Exotic Liquids",
      Country: "UK",
    },
    UnitPrice: 18,
    UnitsInStock: 39,
    UnitsOnOrder: 0,
  },
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f51" },
    CategoryID: 1,
    Discontinued: false,
    ProductID: 2,
    ProductName: "Chang",
    QuantityPerUnit: "24 - 12 oz bottles",
    ReorderLevel: 25,
    Supplier: {
      SupplierID: 1,
      CompanyName: "Exotic Liquids",
      Country: "UK",
    },
    UnitPrice: 19,
    UnitsInStock: 17,
    UnitsOnOrder: 40,
  },
];

```

- dodaj do wyniku inf. o kategoriach

```

db.products.aggregate([
  {
    $match: {},
  },
  //suppliers
  {
    $lookup: {
      from: "suppliers",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Supplier_tmp",
    },
  },
  //categories
  {
    $lookup: {
      from: "categories",
      localField: "CategoryID",
      foreignField: "CategoryID",
      as: "Category_tmp",
    },
  },
  {
    $unwind: "$Supplier_tmp",
  },
  {
    $unwind: "$Category_tmp",
  },
  {
    $addFields: {
      Supplier: {
        SupplierID: "$SupplierID",
        CompanyName: "$Supplier_tmp.CompanyName",
        Country: "$Supplier_tmp.Country",
      },
      Category: {
        CategoryID: "$CategoryID",
        CategorynameName: "$Category_tmp.CategoryName",
      },
    },
  },
  {
    $project: {
      SupplierID: 0,
      Supplier_tmp: 0,
      CategoryID: 0,
      Category_tmp: 0,
    },
  },
  {
    $limit: 2,
  },
]);

```

- wynik

```

[
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f50" },
    Category: {
      CategoryID: 1,
    },
  },
]

```

```

    CategorynameName: "Beverages",
  },
  Discontinued: false,
  ProductID: 1,
  ProductName: "Chai",
  QuantityPerUnit: "10 boxes x 20 bags",
  ReorderLevel: 10,
  Supplier: {
    SupplierID: 1,
    CompanyName: "Exotic Liquids",
    Country: "UK",
  },
  UnitPrice: 18,
  UnitsInStock: 39,
  UnitsOnOrder: 0,
},
{
  _id: { $oid: "63a0607abb3b972d6f4e1f51" },
  Category: {
    CategoryID: 1,
    CategorynameName: "Beverages",
  },
  Discontinued: false,
  ProductID: 2,
  ProductName: "Chang",
  QuantityPerUnit: "24 - 12 oz bottles",
  ReorderLevel: 25,
  Supplier: {
    SupplierID: 1,
    CompanyName: "Exotic Liquids",
    Country: "UK",
  },
  UnitPrice: 19,
  UnitsInStock: 17,
  UnitsOnOrder: 40,
},
];

```

Przykład 2

- dostawcy i produkty

```

db.suppliers.aggregate([
  {
    $match: {},
  },
  //products
  {
    $lookup: {
      from: "products",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Products",
    },
  },
  {
    $addFields: {
      Products: {
        $map: {
          input: "$Products",
          as: "product",
          in: {

```

```

        ProductID: "$$product.ProductID",
        ProductName: "$$product.ProductName",
        UnitPrice: "$$product.UnitPrice",
        Discontinued: "$$product.Discontinued",
    },
},
},
},
{
    $limit: 2,
},
]);

```

- wynik

```

[
  {
    _id: { $oid: "63a05d3cbb3b972d6f4e09f7" },
    Address: "49 Gilbert St.",
    City: "London",
    CompanyName: "Exotic Liquids",
    ContactName: "Charlotte Cooper",
    ContactTitle: "Purchasing Manager",
    Country: "UK",
    Fax: null,
    HomePage: null,
    Phone: "(171) 555-2222",
    PostalCode: "EC1 4SD",
    Products: [
      {
        ProductID: 1,
        ProductName: "Chai",
        UnitPrice: 18,
        Discontinued: false,
      },
      {
        ProductID: 2,
        ProductName: "Chang",
        UnitPrice: 19,
        Discontinued: false,
      },
      {
        ProductID: 3,
        ProductName: "Aniseed Syrup",
        UnitPrice: 10,
        Discontinued: false,
      },
    ],
    Region: null,
    SupplierID: 1,
  },
  {
    _id: { $oid: "63a05d3cbb3b972d6f4e09f8" },
    Address: "P.O. Box 78934",
    City: "New Orleans",
    CompanyName: "New Orleans Cajun Delights",
    ContactName: "Shelley Burke",
    ContactTitle: "Order Administrator",
    Country: "USA",
    Fax: null,
    HomePage: "#CAJUN.HTM#",
    Phone: "(100) 555-4822",
    PostalCode: "70117",
    Products: [

```

```

{
  ProductID: 4,
  ProductName: "Chef Anton's Cajun Seasoning",
  UnitPrice: 22,
  Discontinued: false,
},
{
  ProductID: 5,
  ProductName: "Chef Anton's Gumbo Mix",
  UnitPrice: 21.35,
  Discontinued: true,
},
{
  ProductID: 65,
  ProductName: "Louisiana Fiery Hot Pepper Sauce",
  UnitPrice: 21.05,
  Discontinued: false,
},
{
  ProductID: 66,
  ProductName: "Louisiana Hot Spiced Okra",
  UnitPrice: 17,
  Discontinued: false,
},
},
],
Region: "LA",
SupplierID: 2,
},
];

```

Przykład 3

- zamówienia
- produkty
 - tworzymy kolekcję `products_tmp` (połączenie danych z kolekcji `products`, `suppliers`, `categories`)

```

db.products.aggregate([
  { $match: {} },
  //suppliers
  {
    $lookup: {
      from: "suppliers",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "supplier",
    },
  },
  { $unwind: "$supplier" },
  {
    $project: {
      "supplier._id": 0,
      "supplier.ContactName": 0,
      "supplier.ContactTitle": 0,
      "supplier.City": 0,
      "supplier.Country": 0,
      "supplier.Address": 0,
      "supplier.PostalCode": 0,
      "supplier.Region": 0,
      "supplier.Phone": 0,
      "supplier.Fax": 0,
      "supplier.HomePage": 0,
    }
  }
])

```

```

    },
  },
  //categories
  {
    $lookup: {
      from: "categories",
      localField: "CategoryID",
      foreignField: "CategoryID",
      as: "category",
    },
  },
  { $unwind: "$category" },
  {
    $project: {
      "category._id": 0,
      "category.Description": 0,
      "category.Picture": 0,
    },
  },
  //usunięcie kluczy obcych
  {
    $project: {
      Discontinued: 0,
      QuantityPerUnit: 0,
      ReorderLevel: 0,
      UnitPrice: 0,
      UnitsInStock: 0,
      UnitsOnOrder: 0,
    },
  },
  {
    $project: {
      ProductID: 1,
      ProductName: 1,
      CategoryID: 1,
      CategoryName: "$category.CategoryName",
      SupplierID: 1,
      SupplierName: "$supplier.CompanyName",
    },
  },
  { $out: "products_tmp" },
]);

```

```
db.products_tmp.find();
```

- pozycje zamówień
 - tworzymy kolekcję `orderdetails_tmp` (polecenie wykorzystuje `products_tmp` stworzoną w poprzednim przykładzie)

```

db.orderdetails.aggregate(
  { $match: {} },

  //suppliers
  {
    $lookup: {
      from: "products_tmp",
      localField: "ProductID",
      foreignField: "ProductID",
      as: "product",
    },
  },
  { $unwind: "$product" },
  {
    $project: {

```

```

        "product._id": 0,
    },
},
{
  $project: {
    ProductID: 0,
  },
},
{
  $addFields: {
    TotalValue: {
      $multiply: ["$UnitPrice", "$Quantity", { $subtract: [1, "$Discount"] }],
    },
    Discount: { $round: ["$Discount", 2] },
  },
},
{ $out: "orderdetails_tmp" },
);

```

```
db.orderdetails_tmp.find();
```

- zamówienia
 - każdy dokument agreguje pełną informację o zamówieniu
 - tworzymy kolekcję `orders_tmp` (polecenie wykorzystuje `orderdetails_tmp` stworzoną w poprzednim przykładzie)

```

db.orders.aggregate(
{
  $match: {},
},
//customers
{
  $lookup: {
    from: "customers",
    localField: "CustomerID",
    foreignField: "CustomerID",
    as: "Customer",
  },
},
{
  $unwind: "$Customer",
},
{
  $project: {
    "Customer._id": 0,
    "Customer.ContactName": 0,
    "Customer.ContactTitle": 0,
    "Customer.City": 0,
    // "Customer.Country": 0,
    "Customer.Address": 0,
    "Customer.PostalCode": 0,
    "Customer.Region": 0,
    "Customer.Phone": 0,
    "Customer.Fax": 0,
  },
},
//employees
{
  $lookup: {

```

```

    from: "employees",
    localField: "EmployeeID",
    foreignField: "EmployeeID",
    as: "Employee",
  },
},
{
  $unwind: "$Employee",
},
{
  $project: {
    "Employee._id": 0,
    "Employee.Title": 0,
    "Employee.TitleOfCourtesy": 0,
    "Employee.BirthDate": 0,
    "Employee.HireDate": 0,
    "Employee.Address": 0,
    "Employee.PostalCode": 0,
    "Employee.City": 0,
    "Employee.Region": 0,
    "Employee.Country": 0,
    "Employee.HomePhone": 0,
    "Employee.Extension": 0,
    "Employee.Photo": 0,
    "Employee.Notes": 0,
    "Employee.ReportsTo": 0,
    "Employee.PhotoPath": 0,
  },
},
//shippers
{
  $lookup: {
    from: "shippers",
    localField: "ShipVia",
    foreignField: "ShipperID",
    as: "Shipper",
  },
},
{
  $unwind: "$Shipper",
},
{
  $project: {
    "Shipper._id": 0,
    "Shipper.Phone": 0,
  },
},
//orderdetails
{
  $lookup: {
    from: "orderdetails_tmp",
    localField: "OrderID",
    foreignField: "OrderID",
    as: "Orderdetails",
  },
},
{
  $project: {
    "Orderdetails._id": 0,
    "Orderdetails.OrderID": 0,
  },
},
{
  $project: {
    CustomerID: 0,
    EmployeeID: 0,
    ShipVia: 0,

```

```

    },
  },
  {
    $out: "orders_tmp",
  },
);

```

```
db.orders_tmp.find().limit(2);
```

- wynik

```

[
  {
    _id: { $oid: "63a060b9bb3b972d6f4e1fc6" },
    Customer: {
      CustomerID: "VINET",
      CompanyName: "Vins et alcools Chevalier",
      Country: "France",
    },
    Employee: {
      EmployeeID: 5,
      LastName: "Buchanan",
      FirstName: "Steven",
    },
    Freight: 32.38,
    OrderDate: { $date: "1996-07-04T00:00:00.000Z" },
    OrderID: 10248,
    Orderdetails: [
      {
        UnitPrice: 14,
        Quantity: 12,
        Discount: 0,
        product: {
          ProductID: 11,
          ProductName: "Queso Cabrales",
          QuantityPerUnit: "1 kg pkg.",
        },
        TotalValue: 168,
      },
      {
        UnitPrice: 9.8,
        Quantity: 10,
        Discount: 0,
        product: {
          ProductID: 42,
          ProductName: "Singaporean Hokkien Fried Mee",
          QuantityPerUnit: "32 - 1 kg pkgs.",
        },
        TotalValue: 98,
      },
      {
        UnitPrice: 34.8,
        Quantity: 5,
        Discount: 0,
        product: {
          ProductID: 72,
          ProductName: "Mozzarella di Giovanni",
          QuantityPerUnit: "24 - 200 g pkgs.",
        },
        TotalValue: 174,
      },
    ],
    RequiredDate: { $date: "1996-08-01T00:00:00.000Z" },
  },
]

```

```

ShipAddress: "59 rue de l'Abbaye",
ShipCity: "Reims",
ShipCountry: "France",
ShipName: "Vins et alcools Chevalier",
ShipPostalCode: "51100",
ShipRegion: null,
ShippedDate: { $date: "1996-07-16T00:00:00.000Z" },
Shipper: {
  ShipperID: 3,
  CompanyName: "Federal Shipping",
},
},
{
  _id: { $oid: "63a060b9bb3b972d6f4e1fc7" },
  Customer: {
    CustomerID: "TOMSP",
    CompanyName: "Toms Spezialitäten",
    Country: "Germany",
  },
  Employee: {
    EmployeeID: 6,
    LastName: "Suyama",
    FirstName: "Michael",
  },
  Freight: 11.61,
  OrderDate: { $date: "1996-07-05T00:00:00.000Z" },
  OrderID: 10249,
  Orderdetails: [
    {
      UnitPrice: 18.6,
      Quantity: 9,
      Discount: 0,
      product: {
        ProductID: 14,
        ProductName: "Tofu",
        QuantityPerUnit: "40 - 100 g pkgs.",
      },
      TotalValue: 167.4,
    },
    {
      UnitPrice: 42.4,
      Quantity: 40,
      Discount: 0,
      product: {
        ProductID: 51,
        ProductName: "Manjimup Dried Apples",
        QuantityPerUnit: "50 - 300 g pkgs.",
      },
      TotalValue: 1696,
    },
  ],
  RequiredDate: { $date: "1996-08-16T00:00:00.000Z" },
  ShipAddress: "Luisenstr. 48",
  ShipCity: "Münster",
  ShipCountry: "Germany",
  ShipName: "Toms Spezialitäten",
  ShipPostalCode: "44087",
  ShipRegion: null,
  ShippedDate: { $date: "1996-07-10T00:00:00.000Z" },
  Shipper: {
    ShipperID: 1,
    CompanyName: "Speedy Express",
  },
},
},
];

```